

DAY 06 – IPv4 ASSIGNMENT

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1. What is IPv4?

IPv4 (Internet Protocol version 4) is a network-layer protocol used to identify devices and deliver data packets across IP networks. It uses a 32-bit logical address written in dotted-decimal notation, such as 192.168.1.10.

2. How many bits are in an IPv4 address?

An IPv4 address contains 32 bits. These 32 bits are divided into four groups of 8 bits each.

3. What is an octet?

An octet is a group of 8 bits. In IPv4, each octet can represent a decimal value from 0 to 255.

4. How many octets are in an IPv4 address?

An IPv4 address contains 4 octets. Example: in 192.168.1.10, the octets are 192, 168, 1, and 10.

5. What is a subnet mask?

A subnet mask is a 32-bit value used to separate the network portion of an IP address from the host portion. For example, 255.255.255.0 is commonly used with a /24 network.

6. Difference between network portion and host portion?

The network portion identifies the network/subnet, while the host portion identifies a particular device within that network. The subnet mask determines which bits belong to each portion.

7. What is a private IPv4 address?

A private IPv4 address is an address reserved for use inside private/local networks. It is not directly routable on the public Internet and is commonly used with NAT.

8. Write all three private IPv4 ranges.

The three RFC 1918 private ranges are: 10.0.0.0/8 (10.0.0.0–10.255.255.255); 172.16.0.0/12 (172.16.0.0–172.31.255.255); 192.168.0.0/16 (192.168.0.0–192.168.255.255).

9. What is a public IPv4 address?

A public IPv4 address is an address that is globally routable on the Internet and is normally assigned by an Internet Service Provider or another Internet service. It is different from private RFC 1918 addresses.

10. What is the IPv4 loopback address?

The IPv4 loopback range is 127.0.0.0/8. The most commonly used loopback address is 127.0.0.1, which refers to the local computer itself.

11. What is APIPA?

APIPA stands for Automatic Private IP Addressing. When a device is configured for DHCP but cannot obtain an address from a DHCP server, it may automatically assign itself an address in the 169.254.0.0/16 range.

12. What does 169.254.x.x commonly indicate?

A 169.254.x.x address commonly indicates that the device has assigned itself an APIPA address because it could not obtain a valid IPv4 address from a DHCP server. It can indicate a DHCP/network connectivity problem.

13. What is a default gateway?

A default gateway is the router address a host uses to send traffic to destinations outside its local subnet. On a typical home network, the default gateway is usually the local router.

14. Static vs dynamic IP addressing?

Static IP addressing is manually configured and normally remains fixed until changed. Dynamic IP addressing is automatically assigned, usually by DHCP, and may change when the DHCP lease is renewed or the device reconnects.

15. What is DHCP?

DHCP (Dynamic Host Configuration Protocol) automatically provides network configuration such as an IPv4 address, subnet mask, default gateway, and DNS server to clients.

16. What is a network address?

A network address identifies a subnet rather than an individual host. It is obtained by applying the subnet mask to the IP address. For example, for 192.168.1.25/24, the network address is 192.168.1.0.

17. What is a broadcast address?

A broadcast address is used to send traffic to all hosts on a local subnet. For example, the broadcast address for 192.168.1.0/24 is 192.168.1.255.

18. Why is IPv4 knowledge important in cybersecurity?

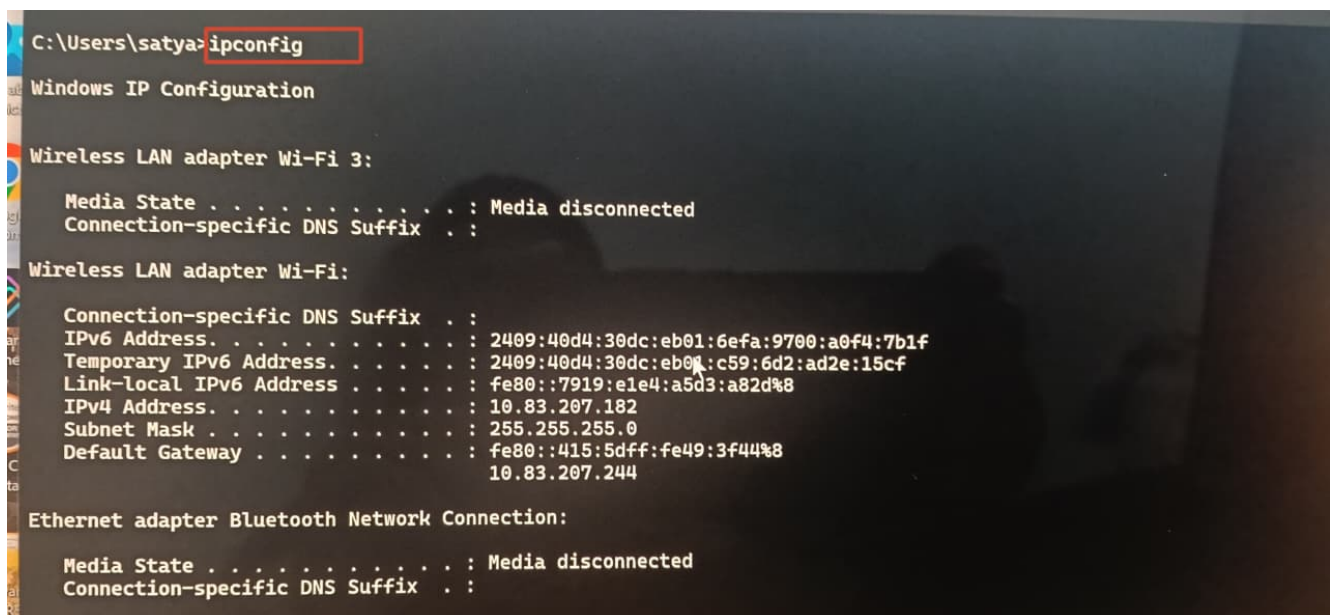
IPv4 knowledge is important for understanding network traffic, identifying hosts, analysing logs, configuring firewalls and IDS/IPS, performing authorised network reconnaissance, troubleshooting connectivity, and investigating suspicious network activity.

Practical Identification

- IPv4:- 10.83.207.182
- Subnet mask:- 255.255.255.0
- Default gateway:- fe80::415:5dff:fe49:3f44%8
- DNS server:- 10.83.217.244
- DHCP :- Enabled

SCREENSHOTS

Ipconfig

A screenshot of a Windows Command Prompt window. The title bar is dark. The command prompt shows the user 'satya' at the C:\Users\ directory. The command 'ipconfig' has been entered and is highlighted with a red rectangular box. The output of the command is displayed below. It shows the configuration for three network adapters: 'Wireless LAN adapter Wi-Fi 3:', 'Wireless LAN adapter Wi-Fi:', and 'Ethernet adapter Bluetooth Network Connection:'. The 'Wi-Fi' adapter is the active one, showing an IPv4 address of 10.83.207.182, a subnet mask of 255.255.255.0, and a default gateway of fe80::415:5dff:fe49:3f44%8. The 'Bluetooth' adapter is also shown with similar information.

```
C:\Users\satya>ipconfig

Windows IP Configuration

Wireless LAN adapter Wi-Fi 3:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . :
    IPv6 Address. . . . . : 2409:40d4:30dc:eb01:6efa:9700:a0f4:7b1f
    Temporary IPv6 Address. . . . . : 2409:40d4:30dc:eb01:c59:6d2:ad2e:15cf
    Link-local IPv6 Address . . . . . : fe80::7919:e1e4:a5d3:a82d%8
    IPv4 Address. . . . . : 10.83.207.182
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : fe80::415:5dff:fe49:3f44%8
                                10.83.207.244

Ethernet adapter Bluetooth Network Connection:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :
```

Ipconfig /all

```
Command Prompt
C:\Users\satya>ipconfig /all

Windows IP Configuration

Host Name . . . . . : Satyam
Primary Dns Suffix . . . . . :
Node Type . . . . . : Mixed
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No

Wireless LAN adapter Wi-Fi 3:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Realtek 8922AE WiFi 7 PCI-E NIC #3
Physical Address. . . . . : 26-B2-B9-86-16-8B
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Wi-Fi 5:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Realtek 8922AE WiFi 7 PCI-E NIC #5
Physical Address. . . . . : 2A-B2-B9-86-16-8B
DHCP Enabled. . . . . : No
Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Wi-Fi:
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Ping 127.0.0.1

```
Command Prompt

DHCP Server . . . . . : 10.83.207.244
DHCPv6 IAID . . . . . : 136622777
DHCPv6 Client DUID. . . . . : 00-01-00-01-30-1C-E9-81-D4-AE-52-5B-90-3A
DNS Servers . . . . . : 10.83.207.244
                        2409:40d4:30dc:eb01::89
NetBIOS over Tcpip. . . . . : Enabled

Ethernet adapter Bluetooth Network Connection:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Bluetooth Device (Personal Area Network)
Physical Address. . . . . : 24-B2-B9-86-16-8C
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes

C:\Users\satya>ping 127.0.0.1

Pinging 127.0.0.1 with 32 bytes of data:
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128

Ping statistics for 127.0.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```